

Impilo vNext Product, Capability and Plane Architecture

Version 2.0 - Controlled Successor Draft

Status: Supersedes vNext V3 v1.2 as a controlling architecture; pending ratification after Target Architecture v1.3.2.

DOCUMENT AUTHORITY

This document is subordinate to the Impilo vNext Hybrid / Federated Target Architecture. Where any conflict exists, the controlling target architecture and its ADRs prevail. This successor draft becomes frozen only after the pending v1.3.2 experience-integrity correction is ratified.

Source lineage: vNext V3 v1.2 (Rewritten Canonical - Production-Grade) + Hybrid / Federated Target Architecture v1.3.1.

Contents

1. Document purpose and authority
 2. Product vision and scope
 3. Architecture mental model
 4. Rings: dependency and maturity, not deployment
 5. Named products and deployment relationship
 6. Capability architecture by plane
 7. Cross-plane product laws
 8. Experience architecture interface
 9. Ecosystem and dual-mode participation
 10. Production readiness and evidence
 11. Product governance
- Appendix A. Legacy V3 disposition register
- Appendix B. Document hierarchy

1. Document purpose and authority

This document defines what Impilo vNext is as a national product: its planes, capability families, named products, ownership boundaries, maturity rings, and reusable platform principles. It replaces the former vNext V3 architectural definition as the authoritative product-and-capability layer.

SUPERSESION NOTICE

The former “Impilo vNext: National Health Operating System - Architectural Definition & Implementation Standard, Version 1.2” is retained as a historical design baseline. It is superseded as a controlling architecture. Content survives only where this successor explicitly retains or amends it.

Document layer	Question answered	Controlling artefact
Target architecture	Where capabilities run, whose authority applies, how nodes federate, how organisations consume services, and how experience is resolved	Hybrid / Federated Target Architecture
Product architecture	What Impilo contains and how capabilities fit together	This document
Technical standards	How APIs, events, trust context, FHIR, federation and operations conform	Technical Standards Catalogue
Domain and experience specifications	How each domain and journey behaves in detail	Experience Completion Packs, clinical packs, regulatory packs and domain specifications
Implementation artefacts	How conformance is expressed in code and operations	ADRs, OpenAPI, AsyncAPI, JSON Schema, migrations, policies, Helm profiles, tests and runbooks

2. Product vision and scope

Impilo vNext is Zimbabwe’s sovereign National Health Operating System: a governed digital public infrastructure for health identity, trust, clinical care, public health, financing, logistics, intelligence, interoperability and user experience. It serves public, private, mission, local-authority, academic, security-sensitive and other accredited participants through one product and multiple governed profiles.

- One national product, not a collection of unrelated apps.
- One codebase and signed artefact set, with declared profile-specific behaviour rather than forks.
- National truth where national authority is required; local operational authority where care is delivered.
- Public-health, payer, clinical and operational capabilities reuse shared primitives rather than rebuilding identity, registries, terminology, audit or workflow foundations.
- Human users experience the platform through intention-led, audience-bound journeys rather than architecture vocabulary.

2.1 Scope boundary

This document does not duplicate federation schemas, trust-domain law, node commissioning, session contracts, detailed user journeys or wire protocols. It defines the capability portfolio and points to the controlling artefact for those concerns.

3. Architecture mental model

The product is organised into seven capability planes. Integration and execution/resilience are cross-cutting frameworks rather than additional business planes. The Experience Plane composes the others but does not become a source of truth for them.

Plane	Mandate	Representative capability family
Kernel - Trust & Truth	Identity, consent, policy, registries, terminology, catalogues, longitudinal record primitives, audit, keys and authority anchors	TSHEPO, VITO, VARAPI, TUSO, INDAWO, MSIKA, ZIBO, BUTANO, UBOMI, MUSHEX and mandatory platform primitives
Clinical - Care Execution	Arrival, encounters, triage, orders, results, referrals, inpatient, theatre, maternity, telemedicine and care pathways	PCT, OROS, diagnostic and departmental modules, telemedicine execution
Public Health, Data & Governance	Surveillance, inspections, complaints, incidents, campaigns, oversight, data stewardship, analytics and research controls	Reusable operational objects and jurisdiction packs
Coverage, Financing & Payer	Membership, benefits, eligibility, contracting, preauthorization, claims, contributions, remittance, settlement and disputes	Coverage services, Ruvimbo surfaces, COSTA/MUSHEX and payer workspaces
Supply, Assets & Logistics	Medicines, stock, procurement, dispatch, devices, equipment and asset operations	Dura, Nhume, Madi and related supply capabilities
Intelligence, Automation & AI	Governed analytics, prediction, summarisation, anomaly detection, recommendation, optimisation and model governance	Model registry, inference gateway, feature views, monitoring and explanation/override
Experience	Public front door, My Life, My Professional, Work, administration, regulation, commissioning, Nompilo, notifications and help	Experience resolver, One UI shell, citizen/provider apps and Experience Completion Packs

3.1 Cross-cutting frameworks

- Integration framework: stable internal and external contracts, FHIR and other standards, developer portal, certification and ecosystem onboarding.
- Execution and resilience framework: node profiles, offline execution, observability, security hardening, support, release governance, backup and disaster recovery.
- Configuration framework: trust domain -> organisation -> node -> facility -> department -> service point, with versioned jurisdiction and capability packs.

4. Rings: dependency and maturity, not deployment

RING RULE

Rings describe capability dependency, stability, evidence and release discipline. Profiles describe deployment. Trust domains describe authority. A Ring 0 component may be national authority, a node-local projection, a local enforcement component or a signed cache without becoming a second National Core.

Ring	Purpose	Representative scope	Exit evidence
Ring 0 - Trust, truth and platform primitives	Establish national and federated authority foundations	Trust, registries, terminology, audit, key management, governed FHIR, schema registry, contract harness, federation identity, finance rail baseline	Policy enforcement, provenance, contracts, audit, backup/restore and compatibility gates proven

Ring	Purpose	Representative scope	Exit evidence
Ring 1 - National care and financing steel threads	Prove end-to-end care and finance execution	PCT, orders/results, referral, scheduling, basic inpatient, eligibility, claims, preauthorisation, remittance and self-service baseline	Care -> record -> eligibility -> claim -> settlement and offline care drills pass
Ring 2 - Scale, public health and intelligence	Expand operations without destabilising core care	Public-health operations, advanced workflow/content, analytics, supply ecosystems, research governance, advanced AI and integrity controls	Independent scaling, governed data access and field-proven workflows

Ring sequencing permits minimum enabling dependencies. Forms, workflow, notification and assistive intelligence needed for a safe Ring 1 journey may be delivered at baseline maturity before their broader Ring 2 platform is complete.

5. Named products and deployment relationship

Named product	Product role	Capability relationship
Impilo National Core	National federation hub and national systems of record	Hosts national authority, ecosystem services, national experiences and federation coordination
Impilo Hospital Node	Federation-autonomous clinical instance for an institution and one or more facilities	Runs local operational care, node trust enforcement and declared local capabilities
Impilo Federation Gateway	Only sanctioned cross-site exchange path	Moves signed, versioned envelopes over mTLS; never raw cross-site Kafka or database replication
Tshepo Local Enforcement Node	Node-local trust plane	Envoy PEP, local PDP/OPA, bundle agent, session context and local audit
Impilo Fleet & Release Service	National node operations plane	Node registry, health, certificates, capabilities, releases and upgrade rings
Impilo Node Bootstrap Agent	Transient signed installer and enrolment agent	Consumes a Bootstrap Manifest; does not create institutional authority
Impilo Facility Edge	Reduced continuity profile for smaller facilities	Local capture, cache and store-and-forward within a declared limited scope

Service-consumption profiles, compute locality, site continuity and trust-domain authority are defined only in the controlling Target Architecture. Product architecture must never infer ownership or access from where software is hosted.

6. Capability architecture by plane

6.1 Kernel - Trust & Truth

Capability	Canonical responsibility	Boundary
TSHEPO trust layer	Identity, authentication, policy decision and enforcement, consent, delegation, context, federation trust, sessions, audit and risk	TSHEPO is a trust layer containing Keycloak, OPA, Envoy, Mvumo and supporting components; it is not one monolithic service
VITO	Client identity, Health ID, deduplication, merge and linkage	Separates identity truth from clinical workflow; cross-domain linkage is governed
VARAPI	Provider identity, standing, qualifications, scope, restrictions and regulatory truth	Employer Work receives only the minimum authorised standing projection

Capability	Canonical responsibility	Boundary
TUSO	Clinical and service-delivery facility registry	Facility identity is stable when deployment routing changes
INDAWO	Non-clinical regulated premises and public-health sites	Do not force premises into the facility model when semantics do not fit
MSIKA	Products, services, orderables, billables, tariffs and benefit references	One service with separately authorised personal, professional and Work surfaces where applicable
ZIBO	Terminology, value sets, maps, confidentiality vocabularies and semantic governance	Versioned bundles support node-local execution
BUTANO	Governed FHIR projections and longitudinal memory	Origin authority remains with the source fact; projections never silently overwrite origin
UBOMI	Births, deaths and civil-registration interface	Reconciliation rather than hidden identity replacement
MUSHEX	Finance switching, authorisation, remittance, settlement and reconciliation rails	Irreversible financial actions require authoritative proof and reversibility
Mandatory primitives	Schema registry, contract tests, audit ledger, KMS/HSM integration, developer portal and AI governance registry	These are operating foundations, not deferred conveniences

6.2 Clinical - Care Execution

- Patient arrival, identity resolution and registration.
- Queue, sorting, triage, emergency, outpatient and virtual care.
- Adaptive clinical encounter cockpit, structured forms, pathways and deterministic decision support.
- Orders, specimens, results, prescribing, dispensing and medication administration.
- Admission, beds, ward rounds, nursing, transfer, discharge, theatre and procedures.
- Maternity, newborn, paediatrics, adult medicine, surgery and specialty pathways.
- Referral, multidisciplinary collaboration, teleconsultation and remote diagnostics.

CLINICAL COMPOSITION PRIORITY

The principal gap is composition, not absence of domain content. The Clinical Work Experience Pack must connect arrival, patient worklists, care setting, cadre logic, form scopes, age-appropriate triage and pathway activation inside the ordinary encounter.

6.3 Public Health, Data & Governance

Public-health and local-authority variation is implemented through shared platform capabilities and versioned jurisdiction packs. City Health is a configured instance of a broader Public Health Operations capability, not a disconnected municipal product.

Canonical operational object	Examples of use
Premises / regulated site	Markets, schools, water points, waste sites, ports, food premises, abattoirs and similar locations
Geographic area / ward / zone / catchment	Jurisdiction, surveillance, deployment and reporting boundaries
Public-health event / complaint / signal	Surveillance, citizen reports, alerts and operational escalation
Inspection / investigation / finding	Environmental health, compliance, laboratory and enforcement workflows

Canonical operational object	Examples of use
Notice / enforcement action	Corrective action, closure, follow-up and legal traceability
Outbreak / incident / response task	Coordination, deployment, logistics, communications and after-action review
Campaign / outreach session	Immunisation, screening, community services and programme operations

6.4 Coverage, Financing & Payer

- Scheme, plan, package and benefit administration.
- Membership, beneficiary enrolment, coverage periods, waiting periods, exclusions and limits.
- Provider contracting, network management and tariff schedules.
- Eligibility, preauthorisation, utilisation review and financial clearance.
- Claims capture, coding edits, adjudication, appeals and integrity review.
- Contributions, invoices, receipts, arrears, subsidies and co-financing.
- Provider payments, remittance, settlement, reversal, reconciliation and recovery.

Financing capabilities consume shared sovereign identity, provider, facility, terminology, clinical, pricing, audit and payment truth. They do not create private copies of the national master data merely because a payer operates a separate workspace.

6.5 Supply, Assets & Logistics

- Medicines, commodities, stock, procurement and distribution.
- Specimen, blood, document, courier and other dispatch workflows.
- Devices, equipment, maintenance, calibration and asset registries.
- Facility-specific integration with eLMIS, analysers, printers, PACS and other local systems.

6.6 Intelligence, Automation & AI

Class	Permitted role	Human control
I1 - Insight only	Dashboards, trends, summaries and anomaly hints	No direct operational effect
I2 - Recommendation	Risk scoring, prioritisation, coding suggestions and decision support	Explicit human acceptance or override
I3 - Governed low-risk automation	Queue routing, reminder prioritisation and low-risk classification	Explicit approval, monitoring, rollback and audit

No AI capability may silently finalise high-risk clinical care, enforcement, entitlement, claims or settlement decisions. Model identity, version, context, timestamp, confidence where relevant and human action must be recorded.

6.7 Experience

- Public experience and care discovery before sign-in.
- My Life: personal health, PHR, consent, coverage, payments, wellness and delegated care.
- My Professional: provider identity, licence, CPD, regulatory self-service, portfolio and invitations.
- Work: active organisation, facility, department, role, shift and operational workflows.
- Organisation, facility, regulatory, node-operations and platform-administration experiences.
- Nompilo guidance, journey persistence, the action centre, help and escalation.

7. Cross-plane product laws

Law	Product rule
Shared truth	Identity, provider, facility, terminology and financial rails are reused wherever lawful; no domain silently creates competing masters.
Module-first	Logical service names do not automatically require separate repositories, deployables or teams. Extraction requires a scaling, resilience, protection, ownership or operational trigger.
Configuration over forks	Jurisdictional, institutional and facility variation uses versioned configuration, forms, workflows, role maps and capability packs.
Clinical safety	Every clinical action declares an authoritative decision and offline/reconciliation posture.
Financial integrity	Money movement, reservation, reimbursement, remittance and settlement are idempotent, auditable, reversible or compensatable, and reconciled.
AI integrity	AI remains assistive by default, governed always, and autonomous only within approved low-risk classes.
Experience honesty	Unavailable, empty, denied, degraded and stale are distinct product states; no false success or silent redirect.
One product	Hospital profiles are declared, tested configurations of one artefact set; independently maintained hospital forks are prohibited.

8. Experience architecture interface

The Product Architecture defines the domains and capability families. The Experience Architecture defines how a person discovers them, enters the correct audience, resumes a journey, understands state and receives help. This document does not independently specify navigation or journey state.

Surface	Product question answered	Controlling specification
Public	What can Impilo help me do before sign-in?	Experience Architecture + Pack P1
My Life	What is happening with my health and what can I do next?	Experience Architecture + Pack P1
My Professional	Am I professionally ready and what requires action?	Experience Architecture + Pack P1
Work	Where am I working and what needs attention now?	Experience Architecture + Pack P3
Organisation / facility / node	How do we claim, configure, commission and operate?	Pack P2
Regulatory / oversight	How do authorities operate complete casework and registers?	Pack P4

EXPERIENCE COMPLETION

Seven packs govern the journey-level work: P1 public/personal/professional; P2 self-service and commissioning; P3 clinical Work; P4 regulatory; P5 accessibility/localisation/content; P6 design system; P7 human usability and field validation.

9. Ecosystem and dual-mode participation

National capabilities are designed for Impilo applications, accredited external applications, institutional nodes, payers, research and approved analytics/AI consumers. “Dual mode” is real only when supported by stable contracts, certification, developer tooling and governed deprecation.

- Developer portal, sandbox, mock services, SDKs and API registration.
- Internal and external API surfaces with explicit audiences and rights.
- Contract and schema compatibility testing for partners.
- Integration profiles for clinical, payer, settlement, public-health and analytics participants.
- Certification, conformance evidence, revocation and lifecycle management.

10. Production readiness and evidence

Evidence family	Minimum proof
Trust and safety	Policy enforcement on the request path, consent and sensitivity controls, break-glass, authority freshness and audit evidence
Federation	Node commissioning, signed exchange, provenance, conflict handling, revocation, reconnection and quarantine
Reliability	SLOs, capacity assumptions, backup/restore, RPO/RTO, disconnection drills and release rollback
Clinical	Arrival-to-discharge workflows, adaptive pathways, orders/results, medication, referral and offline care
Finance	Eligibility, preauthorisation, claim, remittance, settlement, reversal, recovery and reconciliation
Experience	Journey blueprints, web/mobile parity, accessibility, localisation, shared-workstation safety and field validation
Ecosystem	One external integration certified against current contracts and lifecycle rules
AI	One governed workflow with traceable model use, human override, monitoring and rollback

11. Product governance

- Capability ownership is explicit: every canonical object, policy, workflow and projection has one accountable owner.
- The Target Architecture changes only through ADRs and named governance decisions.
- Product capability changes update this document; wire contracts update the Technical Standards Catalogue and machine-readable specifications.
- Journey or visual changes belong to Experience Completion Packs and the versioned Design System unless they alter an architecture invariant.
- Any legacy document marked “canonical” or “frozen” is subordinate once a successor is ratified and must carry a supersession notice.

Appendix A. Legacy V3 disposition register

Legacy section	Disposition	Successor treatment
0. Purpose of rewrite	RETAIN / REFRAME	Retained as historical rationale; successor purpose is product-and-capability governance.

Legacy section	Disposition	Successor treatment
1. Executive summary	AMEND	Retain National Health OS vision; replace National Spine/Sovereign Pod terminology with National Core, trust domains, Hospital Nodes and consumption profiles.
2.1-2.6 capability planes	RETAIN / AMEND	Retain mandates; clarify cross-cutting integration/execution frameworks and current named products.
2.7 Experience Plane	MOVE	Keep only product role and pointer; detailed model belongs to Experience Architecture and packs.
2.8 Public Health reuse	RETAIN	Binding product law.
2.9 Financing reuse	RETAIN	Binding product law.
2.10 AI governance	RETAIN	Binding product law, refined by Technical Standards Catalogue.
3. Architecture laws	DISTRIBUTE	Product-level laws retained here; request context, federation, event and enforcement details move to technical standards and target architecture.
4. Ring 0 service catalogue	RETAIN / UPDATE	Retain capability ownership; correct Tshepo as trust layer and Butano as governed projections.
5. Pod model	SUPERSEDE	Replaced by trust domains, nodes, service profiles, commissioning and federation gateway.
5.1A Jurisdiction packs	RETAIN	Remain versioned configuration artefacts, never forks.
5.2 Federation protocol	MOVE / SUPERSEDE	Product principle retained; protocol moves to Target Architecture and Technical Standards Catalogue.
6. Clinical safety model	RETAIN / AMEND	Keep classes and evidence; “synchronous Kernel” becomes authoritative local or national decision according to trust source.
7. Data/events/read models	DISTRIBUTE	Canonical object catalogues stay here; envelopes and projection rules move to technical standards.
8. Operational architecture	MOVE	Reliability principles remain; implementation standards move to technical catalogue and node profiles.
9. Dual-mode services	RETAIN / MOVE	Product ecosystem principle remains; APIs and certification move to technical standards.
10. Ring strategy	RETAIN / AMEND	Rings are dependency/maturity, never deployment or authority.
11. Component model	RETAIN / UPDATE	Retain module-first discipline and align to current product names.
12. One UI	MOVE	Three domains retained; all navigation, state and journeys governed elsewhere.
13. Production-ready definition	RETAIN / EXPAND	Recast as evidence families aligned to current target acceptance gates.
14. Final position	RETAIN	Impilo is national DPI, not a large app.

Appendix B. Document hierarchy

1. Hybrid / Federated Target Architecture - controlling federation, trust, deployment, experience-model and commissioning architecture.
2. Product, Capability and Plane Architecture - this document; defines product scope and capability ownership.
3. Technical Standards Catalogue - governs machine and operational conformance.
4. Experience Completion Packs and domain specifications - complete journeys and domain behaviour.
5. Machine-readable and operational artefacts - ADRs, APIs, events, schemas, policies, migrations, Helm, tests and runbooks.